

EQUIPMENT INTEGRITY INSPECTION REPORT  
Ref No: IR-2026-08-992A  
Facility: Refinery Unit 3 - Hydrocracker Complex  
Inspection Date: 2026-08-28  
Lead Inspector: J. Doe (ID: MECH-8821)  
Scope: Quarterly Statutory Mechanical Integrity Survey

EXECUTIVE SUMMARY  
This report details the non-destructive testing, visual examination, and vibration diagnostics performed on critical mechanical assets within Hydrocracker Complex Unit 3. Two assets require priority maintenance intervention. One asset exhibits acceptable minor wear within operational tolerance.

ASSET SUMMARY TABLE

| Asset ID   | Type             | Condition Rating | Priority               |
|------------|------------------|------------------|------------------------|
| PUMP-104B  | Centrifugal Pump | CRITICAL         | Priority 1 (Immediate) |
| TANK-301A  | Storage Vessel   | DEGRADED         | Priority 2 (High)      |
| PIPE-EX-12 | Process Piping   | ACCEPTABLE       | Routine Monitor        |

## DETAILED INSPECTION FINDINGS & FIELD MEASUREMENTS

### 1. Asset ID: PUMP-104B (Charge Pump 2B)

- Operational status: Running at 2950 RPM
- Vibration level exceeds ISO standard 10816-3 threshold (7.8 mm/s RMS).
- Inboard bearing housing temperature recorded at 68C (nominal ceiling: 55C).
- Audible cavitation signature present at suction flange.

### 2. Asset ID: TANK-301A (Condensate Storage Vessel)

- Construction: Welded carbon steel, nominal thickness 6.0 mm
- External shell wall thickness measured at 3.1 mm against 4.0 mm nominal minimum.
- Localized pitting corrosion detected on lower shell ring segment 3.
- Foundation settlement within allowable 5 mm limit.

### 3. Asset ID: PIPE-EX-12 (Exchanger Bypass Spool)

- Flange gasket minor weeping observed at Joint #4; within operational tolerance.
- Ultrasonic thickness survey confirms remaining wall thickness of 5.8 mm (min req: 4.2 mm).
- External protective coating intact with no visible atmospheric corrosion.

## INSPECTION CERTIFICATION

I hereby certify that the measurements and visual assessments recorded in this document accurately reflect the physical condition of the assets at the time of inspection.

Signed: J. Doe, Certified Integrity Inspector